Response to Reviewers 1 and 2

Effective Detection of Hardware Trojan Locations in Gate-Level Netlists Using Hybrid GraphSAGE and Cost-Sensitive Ensemble Learning

We thank both reviewers for careful and constructive reports. Reviewer comments are in *italics*; our responses are in blue; every change in the revised manuscript is also blue. Section, table and figure numbers refer to the revised manuscript. **[Done]** means the manuscript has been changed. **Everything in red is still open** and must be resolved before this letter is sent.

Before sending: what the new data change.

1. The five-seed strict-LOCO run gives a macro F_1 of 9.4% for Extra Trees, not the 73.4% in the first submission. The abstract and conclusion now use the new numbers. Tables 3, 5, 6, 7 and Fig. 3 still hold the old single-seed numbers and must be regenerated under the same protocol or clearly relabelled.
2. The B19, wb-conmax and EthernetMAC10GE rows of the old Table 3 are within-family (leave-one-variant-out) results (they match the within-family tables exactly). Please check which protocol produced the *other* rows too (c3540, c5315, s13207 and RS232 also read 100% there, but 2–32% under strict LOCO).
3. Section 8 (c6288 versus c5315) rests on the old numbers; under strict LOCO the order reverses (26.7% versus 2.4%).
4. The RS232 labels look wrong: 22 of 37 netlists have far more Trojan-labelled than benign gates (e.g. 433 versus 11). Your new draft discloses this and blames the alignment against flattened cell libraries, but the labels still need fixing.
5. The correlation and PCA files you sent match every number in your draft (TP–μ_{TP} 0.6301, fan-in–fan-out 0.7123, four components 88.19%, five 93.82%, eighth component 0). But R_v equals $-TP$ exactly, so either your code's "Rare" feature is not the binary indicator $TP < 0.1$ described in Sec. 5.2.2 or TP takes only two values, and the CSV's feature names (TPDiff, NbDist, LZ) do not match the paper's (σ_{TP}, Δ_{TP}, P_0). Check the code against the paper's feature definitions.
6. Your new false-alarm counts are one seed, not the result: the draft's Extra Trees "0 false alarms" on RS232_normal is the best of five seeds (0, 5, 6, 7, 18), and its Random Forest "12" matches no seed (3, 6, 6, 16, 0). The manuscript now uses the five-seed numbers.
7. The jumping-knowledge run (four seeds; seed 999 is missing) shows no consistent benefit. Extra Trees on the hybrid vector: F_1 8.85% with versus 8.12% without JK, within seed noise (per-seed differences +0.2, −0.9, +0.8, +2.8 points); the variant without JK has the higher F_1 in the other five classifier/feature-set pairs. Your draft's "JK delivers higher precision, F_1 and MCC" is true for that one pair only. The manuscript reports the ablation as inconclusive.
8. Run-to-run noise: the raw-feature baseline cannot depend on JK, yet it differs between the five-seed run and the no-JK run (only 579 of 1,284 rows identical; Extra Trees F_1 for seed 7 is 5.83% in one run and 4.49% in the other). Check that the pipeline is deterministic before trusting differences of this size.
9. Table 8's TARMAC column is not TARMAC. Your new draft says it is a 100-threshold sweep over pre-extracted TP values; the real SAT-based TARMAC run takes 2.2 s to 3.03 h (the large-circuit CSV matches your Table 3 exactly). The abstract, conclusion and Sec. 7.5 now say this and no longer claim "2.0 to 6.6 times faster". The "supporting data for A2" is still missing.

Reviewer 1

Main comments

1a. Averaging over variants.

Question. *Table 5 averages over variants, not families (I checked this against Table 3), so RS232, with 32 of 106 variants all at 100%, dominates. Per family, Extra Trees F1 is about 70%, or about 62% without RS232; please report the family-level average as the headline.*

Answer. The headline is now the family-level macro average from the five-seed strict-LOCO runs (Sec. 7.6, Table 9, abstract, conclusion). For Extra Trees on the 23-D vector: balanced accuracy $59.32 \pm 1.67\%$, recall $20.21 \pm 3.00\%$, F_1 $9.40 \pm 2.02\%$, MCC 0.093 ± 0.018, PR-AUC 0.146 ± 0.018 (mean $\pm$ std over five seeds; recomputed from the raw per-circuit results and identical to the supplied summary file). Sec. 6.3 explains that we average over families so that RS232 does not dominate.

Still open.

- regenerate Tables 3, 5, 6, 7 and Fig. 3 under the strict protocol, or relabel them single-seed / within-family as in Sec. 7.1.

1b. Precision when no gate is flagged; s35932.

Question. *Please state how precision is counted when no gate is flagged (0/0). On s35932, Extra Trees has 98.08% accuracy, about 300 errors per variant, with only 45 Trojan nodes. At 7.69% recall, pooled precision should then be near 1%, not 72.83%; Random Forest shows the same pattern, and per-family TP/FP/FN counts would settle it.*

Answer. Precision is computed with `zero_division=0` (Sec. 6.3). For s35932 under strict LOCO, pooled over 35 runs (7 circuits $\times$ 5 seeds, 532,025 gates), Extra Trees flags 125 gates: TP = 36, FP = 89, FN = 10,624, TN = 521,276, i.e. precision 28.8% and recall 0.34%; Random Forest has TP = 36, FP = 66 (Sec. 7.6). Pooled counts for every family, classifier and feature set are in Supplementary Table S1 (CSV). The first submission's counts (45 Trojan nodes for s35932, in Table 2 and behind the old Table 3 row) came from an earlier graph build that has been superseded. The current graphs, with about 303 Trojan gates for s35932, are the correct ones, and Table 2 now reports them. The old per-family results are being regenerated on the current graphs (see 1a), so the old s35932 row (precision 72.83%, recall 7.69%) disappears.

Still open.

- one sentence on what differed between the two builds (labelling definition or graph extraction), in case a reviewer asks; and after regeneration, check that the s35932 Extra Trees row shows precision about 28.8% and recall about 0.34%.

1c. Accuracy.

Question. *A detector that flags nothing already scores 98.5% to 99.99% accuracy here, and on c2670 and s35932 the reported accuracy is lower than that. Please drop the 99.50% accuracy from the abstract and conclusion in favor of MCC, balanced accuracy or PR-AUC; a ranking view (gates inspected to find all Trojan gates) would also help.*

Answer. Accuracy is removed from the abstract and conclusion. Table 9 reports balanced accuracy, MCC and PR-AUC. Sec. 6.3 states that an all-benign predictor scores 98.0–99.99% on every family

except RS232, and that Extra Trees accuracy is below that baseline in 10 of 13 families and equal to it in the other three.

Still open.

- ranking view (gates inspected to find all Trojan gates) not computed; MCC/balanced accuracy/PR-AUC are only in Table 9, not in the old tables.

2a. Dataset description.

Question. *Table 2 omits c3540, c5315 and s13207 despite its "thirteen circuits" caption, and the RS232 (18,500) and AES (52,000) counts look rounded. Trust-Hub's RS232 netlists have only about 200 to 300 gates, and several ISCAS rows match the original benchmark gate counts rather than post-synthesis graphs. Please give exact counts from your own graphs, list all 106 variants, and say how the 5 to 7 TRIT variants per host were chosen.*

Answer. Table 2 now covers all thirteen families with exact post-synthesis node, benign and Trojan counts and Trojan shares, computed from our graphs (107 circuits; 14,309 Trojan gates in 1,442,746 nodes). All circuits with their counts are listed in Supplementary Table S2 (CSV). The seven TRIT-TC variants per ISCAS host (T000–T006) are the complete suite provided by Trust-Hub.

Still open.

- Note: Your new draft now matches the raw file on the circuit counts (107 circuits, 105 infected, AES 5, RS232 Trojan gates 0–556).
- small errors remain in the draft's Table 2: RS232 benign gates are 11–46 in the raw file (not 11–536), and wb_conmax benign gates are 7,608–40,261 (the draft lists only the maximum). The manuscript's Table 2 uses the raw values. The old Table 3 variant counts (AES 16, RS232 32, c3540/c5315/s13207 5 each) still disagree with Table 2 and go away only when Table 3 is regenerated.

2b. Labels through Yosys; Trojan-free circuits.

Question. *Please explain how Trojan labels survive Yosys re-synthesis, and where the Trojan-free circuits sit in the folds; RS232-normal in training would leak the benign part of the test circuit.*

Answer. Sec. 6.1 now explains the two mechanisms (net names containing identifying substrings are preserved; flattened designs are aligned against the synthesized clean netlist) and that Trojan-free circuits sit in the held-out fold of their own family. The two clean controls in our data are RS232_normal and wb_conmax-T300; both sit in the held-out fold of their own family.

Still open.

- RS232 labelling: 22 of 37 netlists (the 90 nm and 180 nm variants) have 416–556 Trojan-labelled and only 11 benign gates. Your draft explains this as an artefact of alignment against flattened standard-cell libraries; if so, the labels are wrong and the family should be relabelled or excluded. Until then RS232 metrics are not comparable; Sec. 6.1 discloses this.

3. Simulation setup.

Question. *All eight behavioral features depend on the simulation, so please describe the simulator, the stimulus, the number of vectors used for TP (is it the 10,000 in Table 8?), and how sequential circuits are reset and clocked.*

Answer. Sec. 6.1 describes the simulator (event-driven, Python, five settling iterations per vector), the stimulus (10,000 vectors for RS232 and AES, 2,000 for B19 and wb-conmax, 1,000 for ISCAS-85/89) and the reset and clocking policy (five reset cycles, clock toggled every step). EthernetMAC10GE uses 2,000 pseudo-random vectors, like B19 and wb-conmax (Sec. 6.1 updated). The Table 8 caption now reads "following tiered feature-extraction logic simulation (10,000 vectors for RS232 and AES, 2,000 for B19, wb-conmax, and EthernetMAC10GE, and 1,000 for ISCAS-85/89)", and the table has a Vectors column. Sec. 7.5 says the proposed-method times are pure model inference (GNN forward pass plus Extra Trees), excluding the one-off logic simulation. All three methods were run on the same tiered datasets, not on a separate 10,000-vector stimulus. Table 8 times exclude logic simulation and graph parsing for all of them: the proposed framework's time is the GNN forward pass plus Extra Trees; TARMAC's is the threshold and clique decision on pre-extracted TP values (hence 0.131 s for AES rather than more than 4 s); Latent AE's is the reconstruction forward pass. The 2.0–6.6× speed-up is therefore an inference-phase speed-up, not end to end, and Sec. 7.5 and the abstract say so. Simulation itself costs 4.06 s for AES-T100 and 0.26 s for RS232-T100 at 10,000 vectors (Supplementary Table S5).

4. Experimental details.

Question. *Please explain how the hyperparameters were chosen (tuning on LOCO test folds would leak), and whether message passing is directed or bidirectional, since the payload sits downstream of the trigger. Please also confirm that feature scaling uses training families only.*

Answer. Sec. 6.4 states that hyperparameters were fixed by 5-fold cross-validation inside the training families of an initial development split and not tuned on held-out families, that edges are directed from driver to load, and that features are standardized per circuit with that circuit's own mean and standard deviation (no cross-circuit statistics). The loss weight $w_{\text{pos}} = N_{\text{benign}}/\max(1, N_{\text{trojan}})$ is recomputed per fold on the training split, as confirmed in the code (Sec. 5.3.1, Table 4).

5. Runtime comparison.

Question. *TARMAC generates test patterns rather than classifying gates, so please state what was timed for each method and on what hardware. About 2 s on EthernetMAC10GE seems low for a SAT-based flow, and "Latent AE" needs a description and a citation. The "better scalability" claim is not supported: Latent AE is faster on 13 of 14 circuits, and the speed-up over TARMAC (2× to 6.6×) does not grow with size. Timings look noisy: B19-T100 and B19-T200 differ by 1.7× at equal size, and AES runs faster than s1423. Section 5.4's "tens of milliseconds" for "the largest evaluated variant, s35932" also conflicts with Table 8, and its complexity analysis omits simulation cost (with max_ depth = None, d is not fixed either).*

Answer. Sec. 7.5 states the hardware (Xeon 2.20 GHz, 16 GB, T4), that times are inference only and exclude logic simulation and graph parsing for all three methods; Latent AE is described (our baseline: two-layer graph autoencoder, 16-D latent, MSE loss). The scalability claim is removed; the text reports 2.0–6.6× over TARMAC, Latent AE faster on 13 of 14 circuits, single-run noise, and Sec. 5.4 now matches Table 8 and notes that d is data-dependent and simulation cost is excluded. Sec. 7.5 now separates the two things. The Table 8 TARMAC column is the online classification step: 100 candidate activation thresholds swept over the pre-extracted TP values in NumPy (1.8 s over 202,839 gates), and the column is marked as such. The complete SAT-based trigger search of our TARMAC re-implementation (Z3 setup plus maximal-clique search, three test patterns, no timeout) is in the new Table 13 and Supplementary Table S8, and runs from 2.2 s (AES-T100) to 10,913 s (s15850, 3.03 h); 785 s for s35932; 56–67 s for EthernetMAC10GE and B19. The abstract, conclusion and Sec. 7.5 no longer claim a speed-up over TARMAC. Latent AE is described as in your draft: one GATConv encoder (to 16 dimensions) and one

GATConv decoder with ELU, 50 epochs of Adam (learning rate 0.01) with MSE on clean benchmark features, with a sweep of 100 threshold percentiles.

Still open.

- (i) The old log (S6) and the new one (S8) agree with each other in kind but not with Table 8, which times a different algorithm, so the TARMAC name over a threshold sweep is misleading; consider renaming it everywhere (including its F_1 values) as a "threshold baseline". (ii) Both baselines sweep 100 thresholds; state whether the best threshold was chosen using the test labels (if so, their F_1 is optimistic and their time includes work the proposed framework does not do). (iii) Your draft's "airtight defence of the 60-second timeout" is not usable: no run used a timeout, and the earlier 60 s statement is out of the manuscript. (iv) On the screenshots: the two Colab runs of s15850 differ by 224 s at 50 signals (801.0 versus 576.9) and 319 s at 100 signals (1,830.7 versus 1,511.7), which is small next to 10,913 s. Z3 runs on the CPU, so this is machine variability and not a GPU speed-up; the first screenshot does not show the hardware.

6. Claims to tone down.

Question. *The method itself uses full-batch training, so full-batch scalability should not be listed as a gap the paper addresses. The introduction calls the evaluation protocol the contribution, "not a new architecture", yet contributions 1 to 3 are architectural. Section 8 is a correlation analysis, not an ablation. Its 50 variants come from only 8 hosts, so $p = 0.026$ is overstated (with $n = 8$ the same r gives $p \approx 0.45$); please analyze at the family level. A near-zero trigger-count correlation also does not "rule out trigger complexity entirely".*

Answer. Full-batch gap removed from the introduction; contributions reworded (components established, combination and evaluation are ours); Sec. 8 renamed "Analysis of the Generalization Gap", described as correlational, with the eight-host caveat ($n = 8$ gives $p \approx 0.45$); trigger-count wording softened. Sec. 8 now also states that it uses the earlier single-seed values and that its c6288/c5315 contrast is not confirmed under strict LOCO.

Suggested additional experiments

E1. Ablation

Question. *Ablation: Please train the trees on the 8 raw features only, the 15-D embedding only, and the full 23-D vector. Three raw features already summarize the neighborhood, and the GNN alone reaches only 48.4% F1, so the embedding's value is unproven. Variants without jumping knowledge, residuals or each weighting stage, plus a feature-importance plot, would add to this.*

Answer. Raw, embedding-only and full vectors are compared for Extra Trees, Random Forest and XGBoost (Tables 9, 10). The full vector is best on balanced accuracy, recall, F_1, MCC and PR-AUC for all three; for Extra Trees recall rises from 5.3% to 20.2% and MCC from 0.045 to 0.093. Absolute performance stays low. Jumping knowledge was ablated over four seeds (Sec. 7.6, Table 12, Supplementary Table S7): removing it changes Extra Trees macro F_1 from 8.9% to 8.1%, within seed variation, and five of six classifier/feature-set pairs score higher without it. The ablation therefore does not show a benefit of JK, and the manuscript says so.

Still open.

- no runs without residual connections or each weighting stage, no feature-importance plot, and seed 999 is missing from the JK run.

E2. Seeds

Question. *Seeds: Please repeat the main runs with at least five seeds and report mean ± standard deviation.*

Answer. Seeds 42, 123, 2024, 7, 999 for the ablation (Table 9). The tree classifiers cannot be ranked: F_1 is 9.4 ± 2.0 (Extra Trees), 8.2 ± 1.4 (Random Forest), 7.8 ± 2.1 (XGBoost).

Still open.

- the older single-seed tables are not repeated.

E3. Out-of-fold embeddings

Question. *Out-of-fold embeddings: The trees train on embeddings from an encoder that already saw those labels; cross-fitting would avoid over-optimistic training embeddings.*

Answer. Not yet addressed in the manuscript.

Still open.

- not run; listed as a limitation.

E4. Prior detectors

Question. *Prior detectors: Please run one or two published gate-level detectors under the same LOCO protocol (e.g., Hasegawa et al.'s features with a random forest, COTD, Mondal et al. [15], or a GNN with public code). Table 1's numbers come from different data and granularity, and the table needs a row for the proposed method.*

Answer. Table 1 now has a row for the proposed method and a granularity column.

Still open.

- no published detector was re-run (your draft only argues that baselines use random splits); run one or keep it as a limitation.

E5. Stimulus sensitivity

Question. *Stimulus sensitivity: Random stimulus may leave parts of large designs idle, so please compare TP features from, say, 1k, 10k and 100k vectors.*

Answer. Run for 12 small circuits (46–2,457 gates) at 1,000 and 10,000 vectors (Sec. 7.5, Supplementary Table S5, CSV). The two circuits with rare nodes (AES_T100: 20; RS232_T100: 21) have the same number at both counts; *TP* correlates at 0.95 and 0.998 for them, but only 0.39–0.68 for the other ten (mean absolute difference 0.009–0.014); simulation time grows 7.8–9.0× for 10× vectors.

Still open.

- your draft overstates this: (i) the CSV has counts of rare nodes, not node sets, so "100% identical" is not shown; (ii) "TP convergence" holds for 2 of 12 circuits, and its own table hides the other seven weak correlations; (iii) the scaling range is 7.78–8.97, not 8.26–8.97; (iv) the statement that b19 and EthernetMAC10GE would take "minutes/hours" has no data; (v) 100,000 vectors, the ISCAS circuits and the large designs were not tested, which is what the reviewer asked.

E6. c6288 TP distributions

Question. *c6288: Benign vs Trojan TP distributions per family (or the share of benign gates with TP < 0.1) would support the multiplier-structure explanation.*

Answer. Not yet addressed in the manuscript.

Still open.

- not run; Sec. 8 calls the topology explanation a hypothesis.

Minor comments

M1. References [17], [18]

Question. *References [17] and [18] need fixing. GNN4Gate is by Dong Cheng, Chen Dong, Wenwu He, Zhenyi Chen and Yi Xu, and [18] pairs BGNN-HT's title (Zhan et al., ISCAS 2023) with the authors of the contrastive GCN paper by Muralidhar et al. (HOST 2021); please split it into two references.*

Answer. GNN4Gate authors corrected; Muralidhar et al. (HOST 2021) is a separate reference. `ref.bib` now lists the BGNN-HT authors from your draft.

Still open.

- Zhan appears only as "J." (full first name needed). Your draft gives pages 13–23 for the Muralidhar HOST 2021 paper; those are the pages of HW2VEC in our bibliography, so they are probably wrong (the entry has no pages for now).

M2. Related work to cite

Question. *Please cite closely related work. The GraphSAGE-based detector in JEIT (doi:10.11999/JEIT221201), GateDet (Cheng et al., 2023), which handles imbalance, and Yasaei et al.'s GCN-based Trojan localization.*

Answer. Not yet addressed in the manuscript.

Still open.

- the draft says the BibTeX entries are in a supplementary references file, which was not sent, so nothing is cited yet. Also check the venue of Yasaei et al.: your draft says HOST 2022, whereas the GCN-based localization paper I know of is in IEEE TVLSI 2022.

M3. GNN4TJ granularity

Question. *GNN4TJ classifies whole RTL designs; it does not identify trigger nodes. Table 1 should show abstraction level and granularity.*

Answer. Introduction corrected; Table 1 has a Granularity column.

M4. Fig. 3

Question. *Fig. 3 misses B19 and wb-conmax, labels AES (about 77% F1) as near-perfect, and its caption says "across models" though only two appear.*

Answer. Caption fixed; text notes the omitted families and that AES is intermediate.

Still open.

- regenerate Fig. 3 with all 13 families.

M5. Fig. 2

Question. *Fig. 2 should show the 8 features feeding GraphSAGE; the result numbers belong in the text.*

Answer. Not yet addressed in the manuscript.

Still open.

- edit `htpipe.pdf`; the figure file was not supplied, so your draft's statement that it was updated cannot be reflected.

M6. Fig. 1 model

Question. *Fig. 1 should state its model ($TP = 2p(1-p)$); the XOR claim holds only for Pb near 0.5.*

Answer. Caption gives $TP = 2p(1-p)$; the XOR claim is qualified.

M7. Equations (1) and (4)

Question. *Equations (1) and (4) are identical.*

Answer. Duplicate removed.

M8. Trigger and payload wording

Question. *"Trigger gates sit dormant until a payload gate is activated" reverses cause and effect, and "dormant payload" (Sec. 5.2.2) conflicts with "active payload" (Sec. 2.2).*

Answer. Rewritten consistently.

M9. R_v threshold

Question. *Rv's fixed 0.1 threshold sits awkwardly with Section 2.2's case against fixed thresholds; a short justification would help.*

Answer. Justified as one of eight input features.

M10. Class-weight wording

Question. *The class weight is "per fold" in Section 5.3.1 but "per circuit" in Table 4, and "inverse-frequency" fits it better than "dynamic". The "instead" in Stage 1 also has no referent.*

Answer. Unified (per-fold, inverse-frequency).

M11. Trojan-free circuits

Question. *Sections 6.3 and 7.4 disagree on whether P/R/F1 are undefined or zero for Trojan-free circuits; false-alarm counts would be more useful there.*

Answer. Both sections say "undefined". Sec. 7.4 gives five-seed false-alarm counts under strict LOCO: wb_conmax-T300 (7,608 gates) has none for any classifier; RS232_normal (46 gates) averages 7.2 false alarms for Extra Trees (0–18), 6.2 for Random Forest (0–16) and 1.2 for XGBoost (0–4). The draft's single-seed figures were replaced by these.

M12. 600 versus 636

Question. *Section 7 says 600 circuit-variant/classifier measurements, but 106 × 6 = 636.*

Answer. Not yet addressed in the manuscript.

M13. Roadmap

Question. *The roadmap in Section 1 skips Section 8.*

Answer. Not yet addressed in the manuscript.

M14. Typos

Question. *Typos: "gold-reference-free", "Graphsage", the incomplete P0 sentence, "detail no comparable study reports" (Sec. 8), stray commas in the author line, and the missing venue of [23] (ICML 2018). Section 7.3 could be titled "Comparison with other classifiers".*

Answer. All listed typos corrected; Sec. 7.3 retitled.

M15. TRIT source and code release

Question. *The 50 TRIT variants come from Trust-Hub, not [27]. Please release the code, graphs, labels and per-variant results publicly rather than "on request"*

Answer. Data availability corrected (13 families, 107 circuits) and promises release of code, graphs, labels and per-variant results at publication.

Still open.

- confirm the release and add the repository URL.

Reviewer 2

R2.1. Number of GraphSAGE layers

Question. *The manuscript presents a hybrid approach for Hardware Trojan (HT) detection, which appears to be a novel and promising concept. The proposed methodology incorporates a three-layer (three-hop) mechanism to identify multi-input triggers converging on a single payload gate. However, the rationale behind selecting exactly three layers requires further justification. The authors are encouraged to provide an experimental comparison involving different hop configurations, ranging from one to n layers, to demonstrate that the three-layer configuration offers an optimal trade-off in terms of detection performance and computational complexity.*

Answer. Sec. 5.2.1 states that three hops is a design choice motivated by multi-input trigger structure and that other depths were not compared; this is a listed limitation. The jumping-knowledge layer keeps the one- and two-hop representations available, but removing it changes the macro F_1 only from 8.9% to 8.1% over four seeds (Sec. 7.6), so we do not claim that jumping knowledge or three hops is optimal. Your new draft now says the same: no layer sweep was run.

Still open.

- the 1–5 layer sweep itself is still not done; it stays a stated limitation.

R2.2. Correlated features

Question. *The proposed methodology employs eight parameters for feature representation. However, some of these parameters appear to be potentially correlated. It would be beneficial if the authors could investigate the possibility of dimensionality reduction by systematically eliminating correlated parameters, either individually or in combinations, and evaluating the resulting performance. Such an analysis would help establish whether all eight parameters are essential and whether the selected eight-dimensional feature representation is indeed the most effective configuration.*

Answer. Sec. 5.2.2 notes that some features are related by construction (R_v is a thresholded TP; μ_{TP}, σ_{TP}, Δ_{TP} summarize the same neighbor values). Sec. 7.6 ablates whole feature groups (raw, embedding, both). Sec. 5.2.2 now reports the measured correlations and PCA (Supplementary Tables S3 and S4, CSV): R_v and TP are perfectly anti-correlated ($r = -1.000$) and the eighth principal component has zero variance; TP and μ_{TP} correlate at 0.63, fan-in and fan-out at 0.71, and the neighborhood-spread feature with fan-in and fan-out at 0.54 and 0.84; four components hold 88.2% of the variance and five hold 93.8%. We keep the original features because the rare-event cue is a threshold on individual features, and state that no PCA-reduced input or per-feature removal was tested.

Still open.

- (a) the number of gates behind the analysis is not stated in the CSVs, and your draft's 359,746 disagrees with the 1,442,746 nodes of the raw results; (b) the CSV feature names do not match the paper's, so confirm what "TPDiff", "NbDist" and "LZ" compute (your draft calls TPDiff Δ_{TP}, but Δ_{TP} as defined in the paper should correlate with degree, as NbDist does); (c) explain why $R_v = -TP$ exactly; (d) the experiments the reviewer asked for, dropping correlated features and retraining, are not run.

R2.3. Meaning of LOCO

Question. *The dual-stage dynamic class imbalance handling mechanism is a notable strength of the proposed approach. The LOCO (Leave-One-Component-Out) evaluation strategy is also convincing and provides a meaningful assessment of the generalization capability of the proposed model.*

Answer. LOCO is Leave-One-Circuit-Out: each fold holds out a whole circuit family. It is now defined at first use in the Introduction.

R2.4. Scalability

Question. *One significant concern relates to the scalability of the proposed approach. The authors have acknowledged that the experiments were conducted using circuits with a relatively small number of nodes, while evaluation on industrial-scale netlists has been deferred to future work. This raises concerns regarding the applicability of the proposed methodology to large and complex circuits. In the Introduction, the authors identify the inability of existing GNN-based detectors to scale effectively to industrial-scale netlists, along with inadequate handling of severe class imbalance, as major limitations of existing techniques. However, the current experimental evaluation does not sufficiently demonstrate how the proposed approach addresses the scalability limitation. Based on the results presented, the applicability of the proposed methodology appears to be established primarily for small- and medium-scale circuits. The authors should acknowledge this limitation more explicitly and provide a clearer discussion of the scalability aspects.*

Answer. Sec. 5.4 has a "Scope of the scalability evidence" paragraph: the largest designs have about 2×10^5 nodes (EthernetMAC10GE 202,839; B19 130,789) with inference under 0.5 s, training uses full-

neighborhood message passing on whole circuit graphs, and the evidence supports small- and medium-scale circuits only. Training and memory are now reported: training is not done per circuit, since the netlists of the 12 training families are batched into composite graphs in each LOCO fold; it takes about 1.5–2.5 minutes per fold with 1.8–3.5 GB peak GPU memory on the T4, while inference is per circuit and takes under 0.5 s (Sec. 5.4). This is repeated in the limitations. Table 11 gives the within-family results for the three largest families with Extra Trees (F1 0.72–1.00), reported honestly: B19-T200 recall is 79.9% and wb-conmax-T100-1 recall is 56.7%, not 100% as your draft states, and all rows use Extra Trees (your draft used Random Forest for B19-T200). Under strict LOCO these families score 0% (Table 10). Per-circuit training time and memory are not available because training is not per circuit; per-fold figures are given instead (see above).

R2.5. Recall and F_1

Question. *Furthermore, considering the critical nature of Hardware Trojan detection, the reported recall and F1-score values require improvement. In particular, a higher recall is desirable to minimize the possibility of undetected Trojans. The authors may investigate additional techniques or suitable modifications to the existing methodology to improve recall while maintaining a reasonable balance between precision and overall detection performance.*

Answer. Sec. 7.2 states that recall is the weak point (macro recall 20.2%, four of five Trojan gates missed) and that a recall-oriented operating point chosen without test data is the main direction. Sec. 7.6 reports best-single-run recall (91.7% for c3540_T000, 83.3% for c6288_T004, 80.9% for AES_T300) together with the five-seed means (56.7%, 56.7%, 68.9%). All reported results use the default decision threshold of 0.5 on every held-out fold, with no tuning on test data (Sec. 6.4). Recall-oriented thresholds (probability 0.20–0.35, or an F_2 criterion chosen on the training folds) are offered only as a deployment recommendation in Secs. 7.2 and 9.

Still open.

- the operating-point experiment itself is not run, so no recall gain is shown.

Additional corrections

(i) Two references to "Figure 3" for the per-family table in Sec. 7.2 now read "Table 3". (ii) A duplicate TrojanSAINT citation key in Sec. 8 was fixed and the duplicate entry removed from `ref.bib`. (iii) Sec. 7.5 has its own heading. (iv) Section 6.1 and Table 2 now use exact graph statistics.